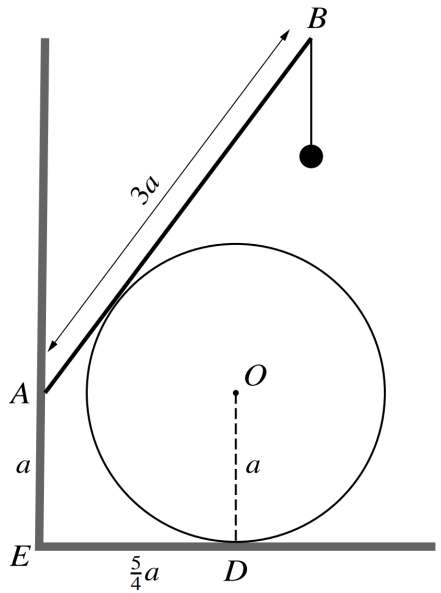
(FM) 2. Equilibrium of a rigid body11

NO. 1



A uniform smooth disc with centre O and radius a is fixed at the point D on a horizontal surface. A uniform rod of length $3a$ and weight W rests on the disc with its end A in contact with a rough vertical wall. The rod and the disc lie in a vertical plane that is perpendicular to the wall. The wall meets the horizontal surface at the point E such that $AE = a$ and $ED = \frac{5}{4}a$. A particle of weight kW is hung from the rod at B (see diagram). The coefficient of friction between the rod and the wall is $\frac{1}{8}$ and the system is in limiting equilibrium. Find the value of k. [8]

A level FURTHER MATHEMATICS 2017/Summer/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
2	Take moments for rod about some point such as: A: $R_P \times A_P - kW \times 3a\cos\theta = W \times (3a/2)\cos\theta$ [$R_P \times 3a/4 - kW \times 9a/5 = W \times 9a/10$ so $15R_P - 36kW = 18W$] O: $F_A \times (5a/4) - kW \times (3a\cos\theta - 5a/4)$ $= -W \times (5a/4 - (3a/2)\cos\theta)$ [$F_A \times 5a/4 - kW \times 11a/20 = -W \times 7a/20$ so $25F_A - 11kW = -7W$] P: $R_A \times AP\sin\theta + F_A \times AP\cos\theta - kW \times (3a - AP)\cos\theta$ $= W \times (3a/2 - AP)\cos\theta$) [$R_A \times 3a/5 + F_A \times 9a/20 - kW \times 27a/20 = W \times 9a/20$ so $12R_A + 9F_A - 27kW = 9W$] B: $R_P \times (3a - AP) - R_A \times 3a\sin\theta - F_A \times 3a\cos\theta$ $= W \times (3a/2)\cos\theta$ [$R_P \times 9a/4 - R_A \times 12a/5 - F_A \times 9a/5 = W \times 9a/10$ so $45R_P - 48R_A - 36F_A = 18W$] C: $R_P \times (3a/2 - AP) - R_A \times (3a/2)\sin\theta - F_A \times (3a/2)\cos\theta + kW \times (3a/2)\cos\theta = 0$ [$R_P \times 3a/4 - R_A \times 6a/5 - F_A \times 9a/10 + kW \times 9a/10 = 0$ so $15R_P - 24R_A - 18F_A + 18kW = 0$] F: $R_P\cos\theta \times (3a - AP)\cos\theta - R_P\sin\theta \times AP\sin\theta - F_A \times 3a\cos\theta = W \times (3a/2)\cos\theta$ [$(3/5)R_P \times 27a/20 - (4/5)R_P \times 3a/5 - F_A \times 9a/5 = W \times 9a/10$ so $81R_P - 48R_P - 180F_A = 90W$]	**M1A1**	F_A here denotes friction on rod measured in downward dirn; P denotes point of contact of rod and disc; θ denotes angle between rod and horizontal. [$AP = 3a/4$, $\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$, $3a - AP = 9a/4$, $3a/2 - AP = 3a/4$] See note below on solving question without introducing R_P (C denotes mid-point of AB) (F is vertically below B, on AO extended)
	Find two more indep. eqns, e.g. resolution of forces on rod: Horizontally: $R_A = R_P\sin\theta$ [$= 4R_P/5$] Vertically: $F_A + (k + 1)W = R_P\cos\theta$ [$= 3R_P/5$]	**B1B1**	A second moment eqn. may be used instead of a resolution

	Along AB: $R_A \cos\theta = F_A \sin\theta + (k + 1)W \sin\theta$ Normal to AB: $R_P = R_A \sin\theta + F_A \cos\theta + (k + 1)W \cos\theta$		Count as 2 eqns if used with moments about P (so R_P absent)
	$F_A = +R_A / 8$ *or* $-R_A / 8$ as appropriate	**B1**	Relate F_A and R_A (may be implied; and must be consistent with friction taken down or up in above eqns)
	[$\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$]	**M1**	Eliminate θ from all reqd. independent eqns. for forces Find either value of k from reqd. independent eqns. for forces
	$F_A \downarrow$: [$R_P = 6W$, $F_A = 3W/5$, $R_A = 24W/5$], $k = 2$ $F_A \uparrow$: [$R_P = 30W/17$, $F_A = 3W/17$, $R_A = 24W/17$], $k = 4/17$	**M1A1**	(or 0.235)
	Total:	**8**	

Explanation

2.

Take moments for rod about some point such as:

A: $R_P \times A_P - kW \times 3a \cos\theta = W \times (3a/2) \cos\theta$

[$R_P \times 3a/4 - kW \times 9a/5 = W \times 9a/10$

so $15R_P - 36kW = 18W$]

O: $F_A \times (5a/4) - kW \times (3a \cos\theta - 5a/4)$

$= - W \times (5a/4 - (3a/2) \cos\theta)$

[$F_A \times 5a/4 - kW \times 11a/20 = - W \times 7a/20$

so $25F_A - 11kW = - 7W$]

P: $R_A \times AP \sin\theta + F_A \times AP \cos\theta - kW \times (3a - AP) \cos\theta$

$= W \times (3a/2 - AP) \cos\theta$)

[$R_A \times 3a/5 + F_A \times 9a/20 - kW \times 27a/20 = W \times 9a/20$

so $12R_A + 9F_A - 27kW = 9W$]

B: $R_P \times (3a - AP) - R_A \times 3a \sin\theta - F_A \times 3a \cos\theta$ M1 A1

$= W \times (3a/2) \cos\theta$

[$R_P \times 9a/4 - R_A \times 12a/5 - F_A \times 9a/5 = W \times 9a/10$

so $45R_P - 48R_A - 36F_A = 18W$]

C: $R_P \times (3a/2 - AP) - R_A \times (3a/2)\sin\theta - F_A \times (3a/2)\cos\theta + kW \times (3a/2)\cos\theta = 0$

[$R_P \times 3a/4 - R_A \times 6a/5 - F_A \times 9a/10 + kW \times 9a/10 = 0$

so $15R_P - 24R_A - 18F_A + 18kW = 0$]

F: $R_P \cos\theta \times (3a - AP)\cos\theta - R_P \sin\theta \times AP \sin\theta$

$- F_A \times 3a \cos\theta = W \times (3a/2)\cos\theta$

$[(3/5)R_P \times 27a/20 - (4/5)R_P \times 3a/5$

$- F_A \times 9a/5 = W \times 9a/10$

so $81R_P - 48R_P - 180F_A = 90W$]

Find two more indep. eqns, e.g. resolution of forces on rod:

Horizontally: $R_A = R_P \sin\theta$ [$= 4R_P/5$]

Vertically: $F_A + (k + 1)W = R_P \cos\theta$ [$= 3R_P/5$]

Along AB: $R_A \cos\theta = F_A \sin\theta + (k + 1)W \sin\theta$ B1 B1 A second moment eqn. may be used instead of a resolution

Normal to AB: $R_P = R_A \sin\theta + F_A \cos\theta + (k + 1)W \cos\theta$

$F_A = +R_A / 8$ or $-R_A / 8$ as appropriate B1 Relate F_A and R_A (may be implied; and must be consistent with friction taken down or up in above eqns)

[$\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$] M1 Eliminate θ from all reqd. independent eqns. for forces Find either value of k from reqd. independent eqns. for forces

$F_A \downarrow$: [$R_P = 6W$, $F_A = 3W/5$, $R_A = 24W/5$], $k = 2$

$F_A \uparrow$: [$R_P = 30W/17$, $F_A = 3W/17$, $R_A = 24W/17$], $k = 4/17$ M1 A1 (or 0.235)

Comment

Candidates are well advised to first identify all the forces acting on the rod, preferably showing them on a diagram so that the symbols used to represent them are clear. This process reveals that three possible forces are unknown, namely the reaction R_A and friction F_A at A and the normal reaction R_P at P, the point of contact of the rod and disc. A further unknown is the required value of k, implying at first sight that four independent equations must be formulated in order to find k. One such follows immediately from the information about limiting equilibrium in the question, namely $F_A = R_A/8$. While there are a variety of possible moment and force resolution equations to choose from, many candidates chose to resolve forces vertically and horizontally and to take moments about A. It is convenient in this process to write the equations in terms of an angle such as OAP, and then replace in due course such factors as the sine and cosine of this angle, namely 4/5 and 3/5 respectively. Some candidates wrongly assumed from the diagram that the radius OP is parallel to DA, resulting in the incorrect values $5/\sqrt{41}$ and $4/\sqrt{41}$ being used for the above sine and cosine. As is sometimes the case in such questions, it is possible to reduce the work required by a careful choice of equations. Thus taking

moments about P, for example, together with only a resolution of forces along AB is sufficient since the unknown force R_P is not introduced. Finally, virtually all candidates gave no indication that the friction at A may act either upwards or downwards, giving different values for k, but full marks could be earned by a satisfactory consideration of only one case.

Answer: 4/17 or 2.

NO. 2

A uniform rod AB has length $2a$ and weight W. The end A rests on rough horizontal ground and the end B rests against a smooth vertical wall. The rod is in a vertical plane that is perpendicular to the wall. The angle between the rod and the horizontal is θ. A particle of weight $5W$ hangs from the rod at the point C, with $AC = xa$, where $0 < x < 1$.

i. By taking moments about A, show that the magnitude of the normal reaction at B is $\frac{W(5x+1)}{2\tan\theta}$. [3]

The particle of weight $5W$ is now moved a distance a up the rod, so that $AC = (x + 1)a$. This results in the magnitude of the normal reaction at B being double its previous value. The system remains in equilibrium with the rod at angle θ with the horizontal.

ii. Show that $x = \frac{4}{5}$. [3]

The coefficient of friction between the rod and the ground is $\frac{2}{3}$.

iii. Given that the rod is about to slip when the particle of weight $5W$ is in its second position, find the value of $\tan\theta$. [5]

A level FURTHER MATHEMATICS 2018/Summer/Paper2_1/Q4

Mark scheme

Question	Answer	Marks	Guidance
4.i	$R_B\ 2a\ \sin\theta = Wa\ \cos\theta + 5Wxa\ \cos\theta$	M1 A1	Moments at A
	$R_B = W\dfrac{(5x+1)}{2\tan\theta}$ **AG**	A1	Simplify to give R_B
		3	
ii	$R_B' = W\dfrac{(5x+5+1)}{2\tan\theta}$ (AEF)	B1	Find new R_B' by moments at A
	$5x + 6 = 2(5x + 1)$, $x = \frac{4}{5}$	M1 A1	Find/verify x from $R_B' = 2R_B$
		3	
iii	$R_A = 6W$, $F_A' = R_B'$ *or* $2R_B[=\dfrac{10W}{2\tan\theta} = 4W]$	B1, B1	Find R_A and new F_A' by resolutions (may be earned earlier)
	$\dfrac{2}{3} = \dfrac{F_A'}{R_A}$	M1	Relate R_A and F_A' using $\mu = \dfrac{F}{R}$
	$= \dfrac{\left(\dfrac{5W}{\tan\theta}\right)}{6W}$ so $\tan\theta = \dfrac{5}{4}$ *or* 1·25	M1 A1	Find $\tan\theta$ **SC** Allow M1 M1 for ground smooth and wall rough (max $\frac{2}{5}$)
		5	

Explanation

4.i

$R_B\ 2a\ \sin\theta = Wa\ \cos\theta + 5Wxa\ \cos\theta$ M1 A1 Moments at A

$R_B = W\dfrac{(5x+1)}{2\tan\theta}$ A1 Simplify to give R_B

ii.

$R_B' = W\dfrac{(5x+5+1)}{2\tan\theta}$ B1 Find new R_B' by moments at A

$5x + 6 = 2(5x + 1)$, $x = \frac{4}{5}$ M1 A1 Find/verify x from $R_B' = 2R_B$

iii.

$R_A = 6W$, $F_A' = R_B'$ or $2R_B\left[=\dfrac{10W}{2\tan\theta} = 4W\right]$ B1 B1 Find R_A and new F_A' by resolutions (may be earned earlier)

$\dfrac{2}{3} = \dfrac{F_A'}{R_A}$ M1 Relate R_A and F_A' using $\mu = \dfrac{F}{R}$

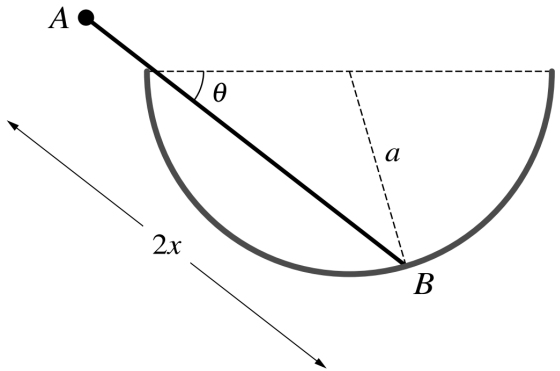
$= \dfrac{\left(\dfrac{5W}{\tan\theta}\right)}{6W}$ so $\tan\theta = \dfrac{5}{4}$ *or* 1·25 M1 A1 Find tanθ

Comment

As in all similar questions, candidates are well advised to first identify all the forces acting on the rod, preferably showing them on a diagram so that the symbols used to represent them are clear. Following the instruction in the question, most candidates succeeded in verifying the given magnitude of the normal reaction at B by taking moments about A. A similar process gives the new magnitude after the movement of the particle up the rod, and using the given information about the ratio of the two magnitudes verifies the given value of x. Finally the magnitudes of the frictional and normal reaction forces at A are found by horizontal and vertical resolution of the forces on the rod when the particle is in its second position, and equating their ratio to the given coefficient of friction gives the required value of tanθ. Taking moments about B, say, is an alternative to a horizontal resolution, but with the disadvantage of more working.

Answers: **(iii)** 1.25.

NO. 3



A uniform rod AB of length $2x$ and weight W rests on the smooth rim of a fixed hemispherical bowl of radius a. The end B of the rod is in contact with the rough inner surface of the bowl. The coefficient of friction between the rod and the bowl at B is $\frac{1}{3}$. A particle of weight $\frac{1}{4}W$ is attached to the end A of the rod. The end B is about to slip upwards when AB is inclined at an angle θ to the horizontal, where $\tan\theta = \frac{3}{4}$ (see diagram).

i. By resolving parallel to the rod, show that the normal component of the reaction of the bowl on the rod at B is $\frac{3}{4}W$. [5]

ii. Find, in terms of W, the reaction between the rod and the smooth rim of the bowl. [4]

iii. Find x in terms of a. [3]

A level FURTHER MATHEMATICS 2019/Summer/Paper2_3/Q5

Mark scheme

Question	Answer	Marks	Guidance
5.i	$R_B \cos\theta + F_B \sin\theta = W \sin\theta + \frac{1}{4} W \sin\theta$	*M1A1	Resolve forces along rod AB ($\sin\theta = 3/5$, $\cos\theta = 4/5$)
	$F_B = \frac{1}{3} R_B$	B1	Relate F_B and R_B (may be implied)
	$(4/5 + \frac{1}{3} \times 3/5) R_B = (1 + \frac{1}{4})(3/5) W$, $R_B = \frac{3}{4}$ W **AG**	DM1 A1	Combine using $\tan\theta = \frac{3}{4}$ to verify R_A
		5	
ii	$R_C = F_B \cos\theta - R_B \sin\theta + W \cos\theta + \frac{1}{4} W \cos\theta$	*M1A1	Resolve forces perpendicular to rod AB, where C denotes rim Substitute to find R_C
	$= (\frac{1}{3} \times \frac{3}{4} \times 4/5) W - (\frac{3}{4} \times 3/5) W + (5/4 \times 4/5) W$	DM1	
	$= (1/5 - 9/20 + 1) W = \frac{3}{4} W$	A1	
		4	
iii	A: $R_C \times AC + (R_B \sin\theta - F_B \cos\theta) \times 2x - W \cos\theta \times x$ $[= W (\frac{3}{4} \times 2x - \frac{3}{4} \times 8a/5 + 9/20 \times 2x - 1/5 \times 2x - 4x/5)]$ C: $(R_B \sin\theta - F_B \cos\theta) \times 2a \cos\theta - W \cos\theta \times CG + \frac{1}{4} W \cos\theta \times AC$ $[= W (9/20 \times 2a - 1/5 \times 2a - 8a/5 + x + \frac{1}{4} \times 2x - \frac{1}{4} \times 8a/5) \cos\theta]$ G: $R_C \times CG - (R_B \sin\theta - F_B \cos\theta) \times x - \frac{1}{4} W \cos\theta \times x$ $[= W (\frac{3}{4} \times 8a/5 - \frac{3}{4} \times x - 9/20 \times x + 1/5 \times x - 1/5 \times x)]$ B: $R_C \times 2a \cos\theta - W \cos\theta \times x - \frac{1}{4} W \cos\theta \times 2x$ $[= W (\frac{3}{4} \times 2a - x - \frac{1}{4} \times 2x) \cos\theta]$	M1 A1	Find total moments about any point, denoting rod's centre by G (taken on either side of C) where $AC = 2x - 2a \cos\theta = 2x - 8a/5$ $CG = x - AC = 2a \cos\theta - x = 8a/5 - x$
	$x = a$	A1	Substitute and equate to zero to find x
		3	

Explanation

5.i

$R_B \cos\theta + F_B \sin\theta = W \sin\theta + \frac{1}{4} W \sin\theta$. M1 A1 Resolve forces along rod AB ($\sin\theta = 3/5$, $\cos\theta = 4/5$)

$F_B = \frac{1}{3} R_B$. B1

$(4/5 + \frac{1}{3} \times 3/5) R_B = (1 + \frac{1}{4})(3/5) W$, $R_B = \frac{3}{4} W$ DM1 A1 Combine using $\tan\theta = \frac{3}{4}$ to verify R_A

ii.

$R_C = F_B \cos\theta - R_B \sin\theta + W \cos\theta + \frac{1}{4} W \cos\theta$. M1 A1 Resolve forces perpendicular to rod AB, where C denotes rim Substitute to find R_C

$= (\frac{1}{3} \times \frac{3}{4} \times 4/5) W - (\frac{3}{4} \times 3/5) W + (5/4 \times 4/5) W$. DM1

$= (1/5 - 9/20 + 1) W = \frac{3}{4} W$. A1

iii.

A: $R_C \times AC + (R_B \sin\theta - F_B \cos\theta) \times 2x - W \cos\theta \times x$
[$=W (\frac{3}{4} \times 2x - \frac{3}{4} \times 8a/5 + 9/20 \times 2x - 1/5 \times 2x - 4x/5)$]

C: $(R_B \sin\theta - F_B \cos\theta) \times 2a \cos\theta - W \cos\theta \times CG + \frac{1}{4} W \cos\theta \times AC$
[$= W (9/20 \times 2a - 1/5 \times 2a - 8a/5 + x + \frac{1}{4} \times 2x - \frac{1}{4} \times 8a/5) \cos\theta$]

G: $R_C \times CG - (R_B \sin\theta - F_B \cos\theta) \times x - \frac{1}{4} W \cos\theta \times x$
[$= W (\frac{3}{4} \times 8a/5 - \frac{3}{4} \times x - 9/20 \times x + 1/5 \times x - 1/5 \times x)$]

B: $R_C \times 2a \cos\theta - W \cos\theta \times x - \frac{1}{4} W \cos\theta \times 2x$
[$= W (\frac{3}{4} \times 2a - x - \frac{1}{4} \times 2x) \cos\theta$]

$x = a$. A1 Substitute and equate to zero to find x

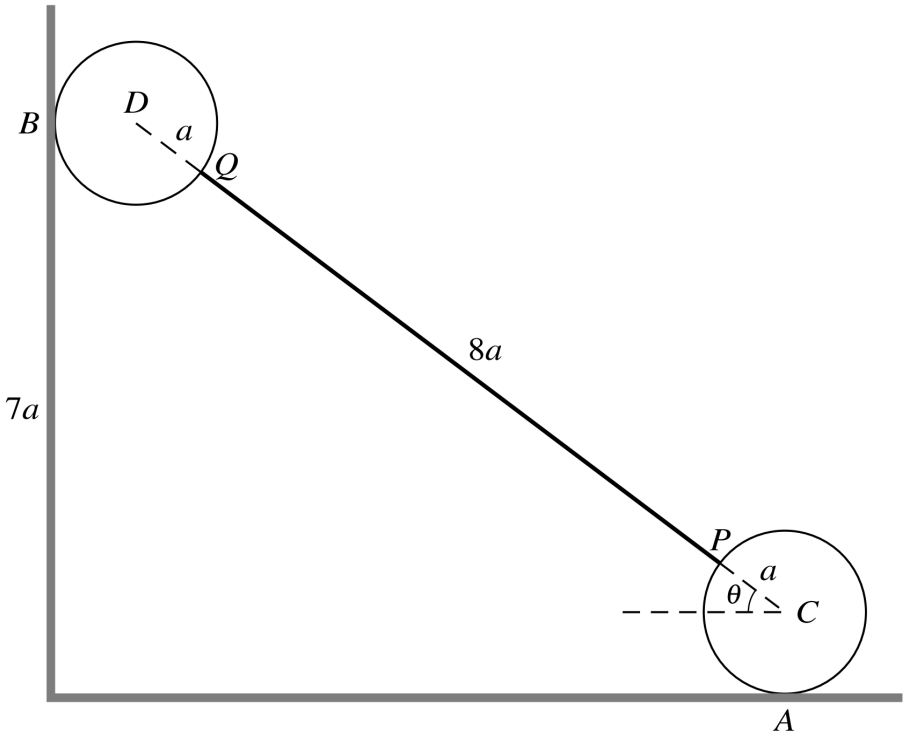
Comment

As in all questions of this type, candidates are well advised to first identify all the forces acting on the rod, preferably showing them on the given diagram, or on their own diagram. This will help the candidate to ensure that they include all the relevant forces when taking moments or resolving. In the first part of the question, candidates were told the method that was to be used to find the normal component of the reaction of the bowl on the rod at B. The instruction 'by resolving parallel to the rod' was designed to help the candidate, by excluding the need to involve the reaction at the rim of the bowl. Those candidates who followed the instruction were usually successful in obtaining the given result. However, many candidates ignored the instruction and took moments about a variety of points and/or resolved horizontally and vertically. As the option to solve the problem 'otherwise' was not offered in the question, candidates should be aware that alternative methods may not receive full credit.

In the second part of the question, any method was acceptable, although the obvious way was to resolve perpendicular to the rod. Those candidates who took moments rarely made any progress, because the distances involved the unknown x.

Finding the distance x needed a moments equation. As always, candidates are well advised to consider an appropriate point about which to take moments, ideally involving as few forces as possible. In this case, B was a good choice, because it involved only W and the reaction force found in the first part of the question.

NO. 4



The end P of a uniform rod PQ, of weight kW and length $8a$, is rigidly attached to a point on the surface of a uniform sphere with centre C, weight W and radius a. The end Q is rigidly attached to a point on the surface of an identical sphere with centre D. The points C, P, Q and D are in a straight line. The object consisting of the rod and two spheres rests with one sphere in contact with a rough horizontal surface, at the point A, and the other sphere in contact with a smooth vertical wall, at the point B. The angle between CD and the horizontal is θ. The point B is at a height of $7a$ above the base of the wall (see diagram). The points A, B, C, D, P and Q are all in the same vertical plane.

i. Show that $\sin\theta = \frac{3}{5}$. [1]

The object is in limiting equilibrium and the coefficient of friction at A is μ.

ii. Find the numerical value of μ. [7]

iii. Given that the resultant force on the object at A is $W\sqrt{(65)}$, show that $k = 5$. [3]

Mark scheme

3.i	Verify sinθ from triangle CDE where E is level with C and vertically below D:	sinθ = 6a / 10a = 3/5 A.G.	**B1**	**[1]**
ii	Resolve forces on object vertically: (may be needed in part (iii) only)	$R_A = (k + 2)W$	**M1 A1**	**[7]**
	Take moments about B : *or* D : *or* A : *or* C : *or* centre of rod:	$F_A\,7a + Wa + kWa(1 + 5\cos\theta) + (W - R_A)a(1 + 10\cos\theta) = 0$ $F_A\,7a + kW5a\cos\theta + (W - R_A)10a\cos\theta = 0$ $R_B\,7a = kW5a\cos\theta + W10a\cos\theta$ $F_A\,a + R_B6a = W10a\cos\theta + kW\,5a\cos\theta$ $F_A\,(5a\sin\theta + a) + R_B5a\sin\theta + W\,5a\cos\theta =$ $R_A\,5a\cos\theta + W5a\cos\theta$	**M1 A1**	
	Find F_A using $R_B = F_A$, sinθ = 3/5, cosθ = 4/5 as necessary: B: D: A or C: Centre of rod:	 $7F_A = 9R_A - 5kW - 10W = 4kW + 8W$ $7F_A = 8R_A - 4kW - 8W = 4kW + 8W$ $7F_A = 4kW + 8W$ $7F_A = 4R_A$ [so R_A is not reqd. here]	**A1**	
	Find μ :	$\mu = F_A / R_A$ = 4/7 or 0·	**M1 A1**	
iii	Equate resultant force at A to $W\sqrt{}(65)$:	$(k + 2)^2 + (4/7)^2(k + 2)^2 = 65$	**M1 A1**	**[3]**
	Solve to verify value of k:	$k^2 + 4k - 45 = (k - 5)(k + 9) = 0$ *or* $(k + 2)^2 = 49$ so $k = 5$ A.G.	**A1**	

Explanation

3.i

sinθ = 6a / 10a = 3/5 A.G. B1 Verify sinθ from triangle CDE

ii.

$R_A = (k + 2)W$ M1 A1 Resolve forces on object vertically:

$F_A\ 7a + Wa + kWa(1 + 5\cos\theta) + (W - R_A)a(1 + 10\cos\theta) = 0$ M1 A1 Take moments about B :

$F_A\ 7a + kW5a\cos\theta + (W - R_A)10a\cos\theta = 0$

$R_B\ 7a = kW5a\cos\theta + W10a\cos\theta$

$F_A\ a + R_B 6a = W10a\cos\theta + kW\ 5a\cos\theta$

$F_A\ (5a\sin\theta + a) + R_B 5a\sin\theta + W\ 5a\cos\theta =$

$R_A\ 5a\cos\theta + W5a\cos\theta$

$7F_A = 9R_A - 5kW - 10W = 4kW + 8W$

$7F_A = 8R_A - 4kW - 8W = 4kW + 8W$ A1 Find F_A using $R_B = F_A$, $\sin\theta = 3/5$, $\cos\theta = 4/5$

$7F_A = 4kW + 8W$

$7F_A = 4R_A$ [so R_A is not reqd. here]

$\mu = F_A / R_A = 4/7$ or 0· M1 A1

iii.

$(k + 2)^2 + (4/7)^2(k + 2)^2 = 65$ M1 A1 Equate resultant force at A to $W\sqrt{(65)}$:

$k^2 + 4k - 45 = (k - 5)(k + 9) = 0$ A1 Solve to verify value of k:

or $(k + 2)^2 = 49$ so $k = 5$ A.G.

Comment

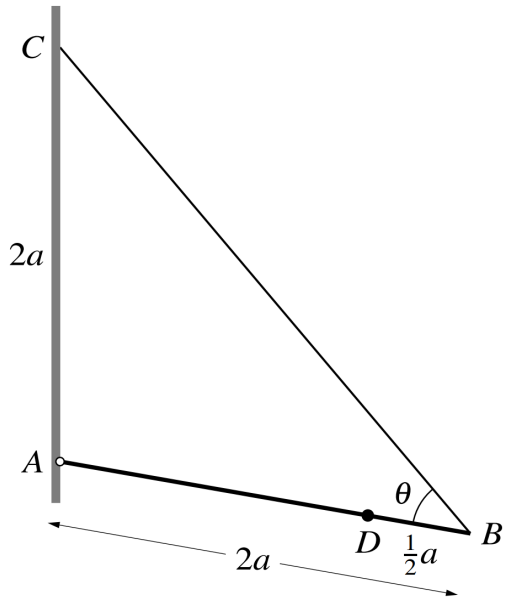
Verifying the given value of $\sin\theta$ from the right-angled triangle with hypotenuse CD was invariably successful. Most candidates then used vertical resolution of forces to find the reaction R_A at A in the form $(k+2)W$, although this is in fact unnecessary in this part if moments are taken about the centre of the rod to give $7F_A = 4R_A$. Most candidates chose instead to take moments for the system about some other point, of which there are several to choose from, in order to find the friction F_A at A in the form $(4/7)(k+2)W$. The numerical value of μ then follows from F_A/R_A. Candidates should not take moments for either sphere about the point P or Q where the rod is attached to the sphere, since the rigid nature of the attachment means that the sum of moments is not necessarily zero. The given value of k may be readily verified in the second part by equating $F_A^2 + R_A^2$ to $65W^2$, provided of course that the correct value of μ has been found.

Answers: **(ii)** 4/7.

NO. 5

Answer only **one** of the following two alternatives.

EITHER



i. The end A of a uniform rod B, of length $2a$ and weight W, is freely hinged to a vertical wall. The end B of the rod is attached to a light elastic string of natural length $\frac{3}{2}a$ and modulus of elasticity $3W$. The other end of the string is attached to the point C on the wall, where C is vertically above A and $AC = 2a$. A particle of weight $2W$ is attached to the rod at the point D, where $DB = \frac{1}{2}a$. The angle ABC is equal to θ (see diagram). Show that $\cos\theta = \frac{3}{4}$ and find the tension in the string in terms of W. [10]

ii. Find the magnitude of the reaction force at the hinge. [4]

Mark scheme

Question	Mark Scheme Details		Part Mark	Total
i	Denoting angle between AB and CA produced by Φ and denoting mid-point of BC by E, say. Take moments for rod about A (AEF):	$W \times a \sin\Phi + 2W \times (3a/2) \sin\Phi = T \times AE$	**M1 A1**	**[10]**
	Substituting $\Phi = 2\theta$ and $AE = 2a \sin\theta$:	$W \times a \sin2\theta + 2W \times (3a/2)\sin2\theta = T \times 2a \sin\theta$	**A1**	
	Hence find tension T in terms of θ:	$T = 2W \sin2\theta / \sin\theta$ [$= 4W \cos\theta$]	**A1**	
	Find T in terms of θ using Hooke's Law:	$T = 3W\,(BC - 3a/2) / (3a/2)$	**M1 A1**	
		$= W\,(8\cos\theta - 3)$	**A1**	
	Equate to find $\cos\theta$:	$4 \cos\theta = 8\cos\theta - 3$	**M1 A1**	
		$\cos\theta = ¾$ **A.G.**	**B1**	
	Hence find T (allow assumption of $\cos\theta = ¾$):	$T = 3W$	**B1**	
ii	Find horizontal force X at A (ignore sign): Find vertical force Y at A (ignore sign):	$X = T \sin\theta$ [$= 3W\sqrt{7} / 4$] $Y = 3W - T \cos\theta$ [$= 3W / 4$]	**B1**	**[4]**
	Find magnitude of force at A:	$\sqrt{(X^2 + Y^2)} = (3/\sqrt{2})\ W$ *or* $(3\sqrt{2} / 2)W$ *or* $2.12\ W$	**M1 A1**	

Explanation

i

W × a sinΦ + 2W × (3a/2) sinΦ = T × AE M1 A1 Take moments for rod about A (AEF):

W × a sin2θ + 2W × (3a/2)sin2θ = T×2a sinθ. A1 Substituting Φ = 2θ and AE = 2a sinθ:

T = 2W sin2θ / sinθ [= 4W cosθ]. A1 Hence find tension T in terms of θ:

$T = 3W\,(BC - 3a/2) / (3a/2)$ M1 A1 Find T in terms of θ using Hooke's Law:

$= W\,(8\cos\theta - 3)$

$4\cos\theta = 8\cos\theta - 3$ M1 A1 Equate to find $\cos\theta$:

$\cos\theta = ¾$ A.G.

$T = 3W$

ii.

$x = T\sin\theta\ [= 3W\sqrt{7} / 4]$ B1 Find horizontal force X at A (ignore sign):

$Y = 3W - T\cos\theta\ [= 3W / 4]$. Find vertical force Y at A (ignore sign):

$\sqrt{(X^2 + Y^2)} = (3/\sqrt{2})\,W$ M1 A1 Find magnitude of force at A:

or $(3\sqrt{2} / 2)W$ or 2.12 W

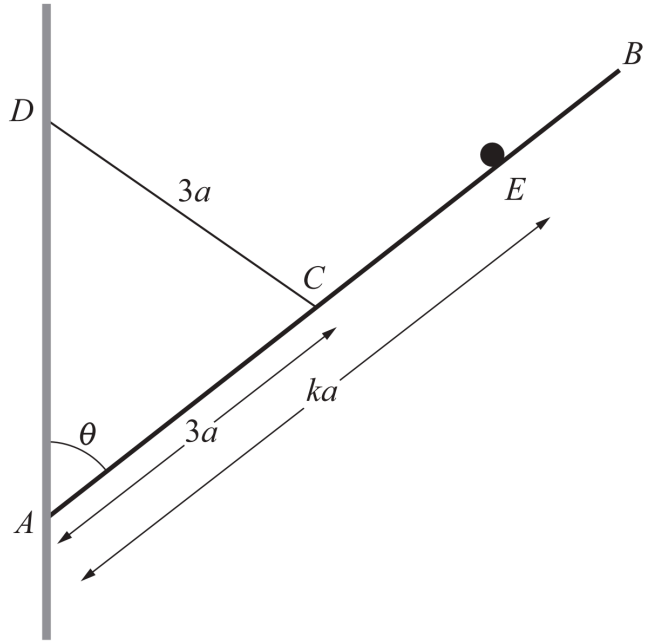
Comment

(Mechanics)

This optional question was attempted by only 15% of the candidates, but many of those who did so made good attempts. The tension T in the string in terms of θ may be found in two ways, namely by taking moments for the rod about A and by applying Hooke's law to the string. Equating the two resulting expressions for T verifies the given value of $\cos\theta$ and yields the required value of T. The horizontal and vertical components of the reaction force at the hinge may be found by corresponding resolution of forces, and then combined in the usual way to give the required magnitude.

Answer: $3W$; $(3/\sqrt{2})W$ or $2.12W$.

NO. 6



The end A of a uniform rod AB of length $6a$ and weight W is in contact with a rough vertical wall. One end of a light inextensible string of length $3a$ is attached to the midpoint C of the rod. The other end of the string is attached to a point D on the wall, vertically above A. The rod is in equilibrium when the angle between the rod and the wall is θ, where $\tan\theta = \frac{3}{2}$. A particle of weight W is attached to the point E on the rod, where the distance AE is equal to ka $(3 < k < 6)$ (see diagram). The rod and the string are in a vertical plane perpendicular to the wall. The coefficient of friction between the rod and the wall is $\frac{1}{3}$. The rod is about to slip down the wall.

a. Find the value of k. [5]

b. Find, in terms of W, the magnitude of the frictional force between the rod and the wall. [2]

A level FURTHER MATHEMATICS 2024/Winter/Paper3_2/Q4

Mark scheme

Question	Answer	Marks	Guidance
a	In this question, allow equivalent marks for resolutions in different directions and moments about other points. Apply the guidance given in the main scheme. $\uparrow T\cos\theta + F = 2W$	B1	
	$\rightarrow T\sin\theta = R$	B1	
	Moments about A: $T\cos\theta \times 3a\sin\theta + T\sin\theta \times 3a\cos\theta = W\times 3a\sin\theta + W \times ka\sin\theta$ OR Moments about C: $R \times 3a\cos\theta = F \times 3a\sin\theta + W\times(ka - 3a)\sin\theta$	M1	All relevant terms included, dimensionally correct, forces must be resolved if appropriate. Allow sin/cos mix, allow sign errors. LHS: any equivalent expression, for example $3Ta\sin 2\theta$, $3Ta\sin(180 - 2\theta)$. All relevant terms included, dimensionally correct, forces must be resolved if appropriate. Allow sin/cos mix, allow sign errors.
	[$6aT\cos\theta = (3+k)aW$ and $\frac{T}{3}(3\cos\theta+\sin\theta)=2W$ give] [*and giveand give*] $12\cos\theta=(3+k)(\cos\theta+\frac{1}{3}\sin\theta)$	M1	Use $F=\frac{1}{3}R$ and eliminate T and W to obtain an expression in k and θ, dependent on a dimensionally correct moments equation.
	k =5	A1	
		5	
b	A complete method to find F in terms of W	M1	Any complete method to find F. For example, substitute into moments equation to obtain T [$T=\frac{2}{3}\sqrt{13}\ W,\ R=2W$]

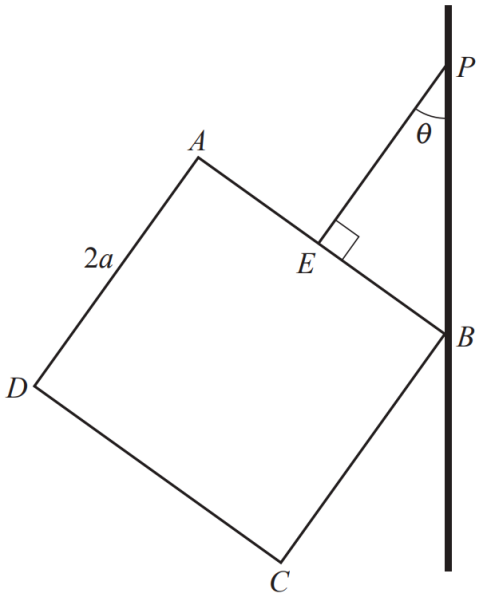
	$F = \frac{2}{3}W$	A1	Correct.
		2	

Comment

a. The majority of candidates made a reasonable attempt at answering this part. Almost all these candidates were able to resolve vertically and horizontally for the system and obtain two correct equations connecting the forces. These candidates also knew that they needed a moments equation. Most candidates took moments about the point *A*, so that the forces at *A* were not involved. However, many had difficulty in finding the moment of the tension, often omitting either the vertical or the horizontal component of the tension.
Some candidates took moments about *C* and this approach was usually more successful in including all the relevant terms.
Having written down three equations (two resolution and one moments), most of these candidates were able to find a value for *k*.

b. This part was usually answered well by candidates.

NO. 7



A uniform square lamina of side $2a$ and weight W is suspended from a light inextensible string attached to the midpoint E of the side AB. The other end of the string is attached to a fixed point P on a rough vertical wall. The vertex B of the lamina is in contact with the wall. The string EP is perpendicular to the side AB and makes an angle θ with the wall (see diagram). The string and the lamina are in a vertical plane perpendicular to the wall. The coefficient of friction between the wall and the lamina is $\frac{1}{2}$.

Given that the vertex B is about to slip up the wall, find the value of $\tan\theta$. [8]

A level FURTHER MATHEMATICS 2023/Winter/Paper3_1/Q3

Mark scheme

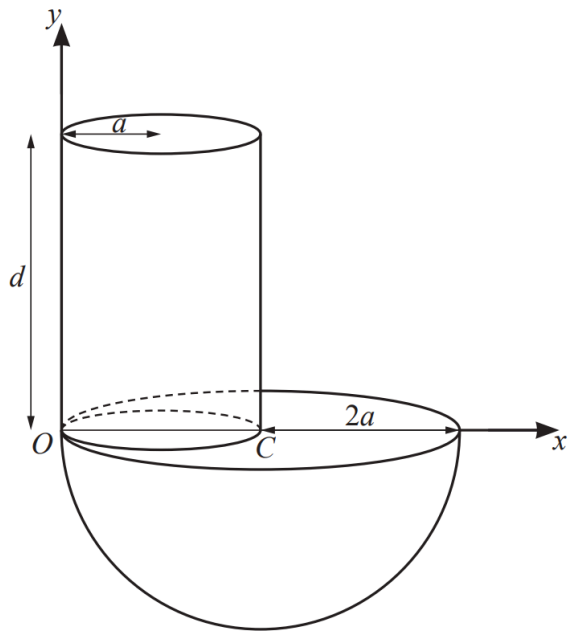
Question	Answer	Marks	Guidance
3	Let N be normal reaction at B and F the frictional force acting downwards $\uparrow T\cos\theta = F + W$	B1	
	$\rightarrow T\sin\theta = N$	B1	
	Moments about B: $Ta = W\sin\theta \times a + W\cos\theta \times a$	M1A1	A moments equation with all relevant forces.
	$F = \frac{1}{2}N$ used	M1	
	$\left(\cos\theta + \sin\theta\right)\left(\cos\theta - \frac{1}{2}\sin\theta\right) = 1$ oe	M1	Combine to obtain equation in θ. Equation in trigonometric functions only.
	$\frac{1}{2}\cos\theta\sin\theta = \frac{3}{2}\left(\sin\theta\right)^2$ $\sin\theta\left(\cos\theta - 3\sin\theta\right) = 0$	M1	Solve trigonometric equation.
	$\tan\theta = \frac{1}{3}$	A1	
		8	

Comment

This question proved challenging for many candidates, who nevertheless often scored the marks allocated to the equilibrium of forces, including the use of the equation $F = \mu R$. Stronger candidates took moments about point P and quickly obtained the correct answer, often in an elegant way. In questions where many forces are involved candidates are encouraged to examine the diagram carefully before selecting the point about which to take moments. A good choice can greatly simplify the subsequent working, thus decreasing the chance of making errors as well as the time spent on calculations. The best answers included a fully annotated diagram, with an indication of the letters used to represent the forces active on the system. Weak candidates did not annotate the

diagram and sometimes used letters that are not usually associated with such forces, e.g. X for the normal reaction, instead of N or R .

NO. 8



An object is formed from a solid hemisphere, of radius $2a$, and a solid cylinder, of radius a and height d. The hemisphere and the cylinder are made of the same material. The cylinder is attached to the plane face of the hemisphere. The line OC forms a diameter of the base of the cylinder, where C is the centre of the plane face of the hemisphere and O is common to both circumferences (see diagram). Relative to axes through O, parallel and perpendicular to OC as shown, the centre of mass of the object is $(\overline{x},\overline{y})$.

a. Show that $\overline{x} = \dfrac{32a^2 + 3ad}{16a + 3d}$ and find an expression, in terms of a and d, for $\overline{y}$. [5]

The object is placed on a rough plane which is inclined to the horizontal at an angle θ where $\sin\theta = \frac{1}{6}$. The object is in equilibrium with CO horizontal, where CO lies in a vertical plane through a line of greatest slope.

b. Find d in terms of a. [3]

Mark scheme

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="5">a</td><td>[Mass is proportional to volume]
<table>
<tr><td></td><td>Volume</td><td>Distance of centre of mass from vertical axis</td><td>Distance of centre of mass from OC</td></tr>
<tr><td>Hemisphere</td><td>$\frac{2}{3}\pi(2a)^3$</td><td>$2a$</td><td>$-\frac{3}{8}\times 2a$</td></tr>
<tr><td>Cylinder</td><td>$\pi a^2 d$</td><td>a</td><td>$\frac{1}{2}d$</td></tr>
<tr><td>Object</td><td>$\frac{2}{3}\pi(2a)^3+\pi a^2 d$</td><td>$\bar{x}$</td><td>$\bar{y}$</td></tr>
</table>
$\left(\frac{2}{3}\pi(2a)^3+\pi a^2 d\right)\bar{x}=\frac{16}{3}\pi a^3\times 2a+\pi a^2 d\times a$</td><td>M1 A1</td><td>Moments equation, dimensionally correct, correct number of terms. Allow sign errors.</td></tr>
<tr><td>Simplify to $\bar{x}=\frac{32a^2+3ad}{16a+3d}$</td><td>A1</td><td>AG. At least one line of intermediate working.</td></tr>
<tr><td>$\left(\frac{2}{3}\pi(2a)^3+\pi a^2 d\right)\bar{y}=\frac{16}{3}\pi a^3\times\left(-\frac{3}{8}\times 2a\right)+\pi a^2 d\times\frac{1}{2}d$</td><td>M1</td><td>Moments equation, dimensionally correct, correct number of terms. Allow sign errors.</td></tr>
<tr><td>$\bar{y}=\frac{3\left(d^2-8a^2\right)}{2(16a+3d)}$</td><td>A1</td><td>AEF</td></tr>
<tr><td></td><td>5</td><td></td></tr>
<tr><td>b</td><td>$\sin\theta=\frac{2a-\bar{x}}{2a}$</td><td>B1</td><td></td></tr>
</table>

	$2a \times \frac{1}{6} = 2a - \frac{32a^2 + 3ad}{16a + 3d}$ $\frac{5}{3}(16a + 3d) = (32a + 3d)$	M1	Remove fractions
	$d = \frac{8}{3}a$	A1	
		3	

Comment

a. In this part question one of the answers was given, therefore the candidates were expected to show their working with particular care to clearly communicate their understanding. The best answers involved the use of tables with volumes (as mass is proportional to volume here) and distances of the centre of mass for the hemisphere, the cylinder, and the object all explicitly stated. Many good responses were seen to this question, with most realising that the centre of mass of the hemisphere was below the *x*-axis. Some responses included equations that were not dimensionally correct. Candidates should be encouraged to check that the equations they write are dimensionally consistent.

b. This part question proved challenging, with few responses seen that were fully correct. Again, the most successful responses made use of a suitable diagram. By drawing such a diagram, it could be seen that application of the tangent ratio was not appropriate here. The better responses made use of the hint that $\sin(\theta) = \frac{1}{6}$ and hence applied the sine ratio instead, often with good results.